

The discretization cliff: clique blow-ups of the 5-cycle and a predictable failure mode of database-verified conjecture generators

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Abstract

Machine conjecture generators in the Graffiti lineage — Fajtlowicz’s *Graffiti*, DeLaViña’s *Graffiti.pc*, Larson and Van Cleemput’s *CONJECTURING*, Davila’s *TxGraffiti* — emit an inequality only after it has survived evaluation on a finite stored database of graphs, and typically only if the database also contains a graph attaining it with equality. We isolate a structural regime in which this filter is systematically uninformative. On a clique blow-up $H[K_m]$, every “induced-subgraph number” associated with a hereditary class of bipartite graphs — the forest number f , the bipartite number b , the induced-tree number $tree$, the induced-path number $path$ — is *pinned*: it is bounded by $2\alpha(H)$ for all m and constant in m for $m \geq 2$, while n , Δ , and the distance-, parity- and density-derived correction terms of the same invariant vocabulary grow with m . Any conjectured lower bound on a pinned invariant by a growing term therefore fails for all large m , and whenever the growing term lands exactly on the pinned value, the last member on which the bound survives does so with *equality*.

We work this out completely for $H = C_5$. For every $m \geq 1$ we prove $\alpha(C_5[K_m]) = 2$ and $path = tree = f = b = 4$, and we give closed forms for the growing side ($\Delta = 3m - 1$, $dist_{\text{even}} = 2m + 1$, $hor_{\text{even}} = m(2m - 1)$, $n \bmod \Delta = 2m + 1$ for $m \geq 3$). Four conjectures of *Written on the Wall II* that were open in July 2026 — numbers 63, 85, 64 and 309 — are then refuted by the family with exact thresholds: all three of 63, 85, 64 are attained with equality at $m = 3$ (order 15) and violated for every $m \geq 4$; 309 is attained with equality at $m = 1$ and violated for every $m \geq 3$. An exhaustive evaluation of all 522 transcribed WOWII records against $C_5[K_4]$ locates the same mechanism eight more times among conjectures already refuted by others, and — the larger half of the data — exhibits the family as a zero-slack witness for roughly three dozen further statements, including 19 theorems of the corpus. We state explicitly what this does *not* establish: it is a mechanism and a diagnostic, not a counterexample engine, and we make no claim here that tightness data can be navigated prospectively.

1 Introduction

1.1 Database-verified conjecture generation

A conjecture generator of the Graffiti type maintains a finite collection of graphs $\mathcal{D}$ together with precomputed values of a fixed vocabulary of invariants, forms candidate inequalities between algebraic expressions in that vocabulary, and discards every candidate falsified by some member of $\mathcal{D}$ [11, 6, 7, 16, 17, 4]. Surviving candidates are further filtered by significance heuristics; in Fajtlowicz’s design and its descendants the dominant heuristic (the “Dalmatian” rule and its variants) prefers bounds that are *attained* on $\mathcal{D}$ and that are not dominated by an already-retained bound [17]. The output is thus, by construction, a set of statements that are true on $\mathcal{D}$ and sharp on $\mathcal{D}$.

This is a strong filter, and the record of the programs shows it works: many Graffiti and Graffiti.pc conjectures have become theorems, and DeLaViña’s *Written on the Wall II* (WOWII) list [8] remains an actively worked source of open problems two decades after its first entries. But the filter has an exact blind spot, and the blind spot is not random. It is the set of graph families along which some invariants of the vocabulary are *constant* while others are *unbounded*. If $\mathcal{D}$ contains the small members of such a family and not the large ones, then a candidate bound relating a constant invariant to an unbounded one

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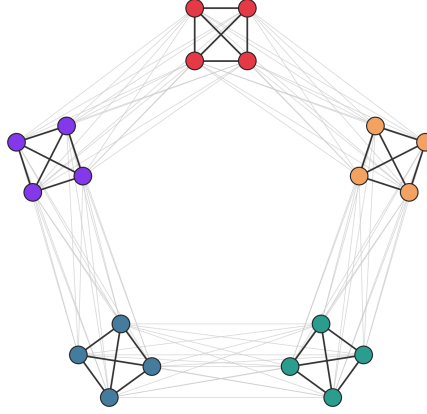


Figure 1: $C_5[K_4]$: five K_4 blobs placed on a 5-cycle, with consecutive blobs completely joined. It has 20 vertices, 110 edges, is 11-regular, vertex-transitive, and has diameter 2.

is not merely unfalsified on $\mathcal{D}$ — it is likely to be *tight* on $\mathcal{D}$, and therefore promoted by the significance heuristic rather than discarded by it. A conjecture that reaches the published list this way is false, and the very evidence that made it look good — sharpness at the top of the database — is the signature of its falsity.

1.2 Contribution

This paper does three things.

1. **States the mechanism generally and proves the pinning half of it** (Section 3). For a clique blow-up $G = H[K_m]$ we prove that $\alpha(G) = \alpha(H)$ for all m , that $\text{path}(G) \leq \text{tree}(G) \leq f(G) \leq b(G) \leq 2\alpha(H)$ for all m , and that the exact value of any hereditary-bipartite-class induced-subgraph number of G is determined by a profile formula over $V(H)$ that does not mention m once $m \geq 2$ (Lemma 3.1, Theorem 3.2, Proposition 3.5). We characterise when the pinning is at the ceiling: if $\text{path}(H) = 2\alpha(H)$ then all four numbers equal $2\alpha(H)$ for every $m \geq 1$. Odd cycles satisfy this hypothesis, so $C_{2r+1}[K_m]$ pins all four numbers at $2r$ for every m (Corollary 3.4).
2. **Carries out the analysis of $C_5[K_m]$ in closed form and derives four cliffs** (Sections 4–5). Each of WOWII 63, 85, 64 and 309 is shown to reduce on this family to an explicit arithmetic inequality in m , with an exact threshold. Three of the four have their last surviving member at $m = 3$, with equality, and fail for every $m \geq 4$.
3. **Reports an exhaustive empirical evaluation of the carrier against the whole WOWII corpus** (522 transcribed records; Section 7), with the corpus’s own defects stated rather than exploited, and argues that the sharpness data — roughly three dozen zero-slack statements, 19 of them proved theorems — is the more informative half of the result.

Section 8 states what is not claimed. In particular, an obvious follow-up question — whether the tightness pattern can be used *prospectively*, to steer a search towards the next counterexample — is raised but explicitly not answered here; the evidence available to us is anecdotal and we say so.

1.3 The carrier

The graph of Figure 1 was found on 2026-07-23 while searching WOWII for a counterexample to Conjecture 85. It turned out to refute Conjectures 63 and 85 simultaneously; Gebendorfer subsequently used the same carrier family to refute 309 [14] and 64 [13], in both cases crediting the earlier certificate. Four open conjectures fell to one 20-vertex graph and its siblings. The point of this paper is not the count but the explanation: the four kills are four instances of a single arithmetic accident that the generator’s own verification procedure could not have detected.

2 Preliminaries

All graphs are finite, simple and undirected. For a graph G we write $V(G)$, $E(G)$, $n(G) = |V(G)|$, $\Delta(G)$ and $\delta(G)$ for the maximum and minimum degree, $d_G(u, v)$ for the distance, $\bar{G}$ for the complement, and $G[S]$ for the subgraph induced by $S \subseteq V(G)$. We use $\alpha(G)$ for the independence number, $\omega(G)$ for the clique number, $\gamma(G)$, $\gamma_t(G)$, $\gamma_c(G)$ for the domination, total domination and connected domination numbers, and $L_s(G)$ for the maximum number of leaves of a spanning tree, so that $L_s(G) = n(G) - \gamma_c(G)$ for connected G with $n \geq 3$.

The four “induced-subgraph numbers” that carry the argument are, in DeLaViña’s notation [9]:

$$\begin{aligned} f(G) &= \max\{|S| : G[S] \text{ is a forest}\}, \\ b(G) &= \max\{|S| : G[S] \text{ is bipartite}\}, \\ \text{tree}(G) &= \max\{|S| : G[S] \text{ is a tree}\}, \\ \text{path}(G) &= \max\{|S| : G[S] \text{ is a path}\}. \end{aligned}$$

We also use two parity invariants of the WOWII vocabulary. For $v \in V(G)$,

$$\begin{aligned} \text{dist}_{\text{even}}(v) &= |\{x \in V(G) : d_G(v, x) \text{ is even}\}|, \\ \text{hor}_{\text{even}}(v) &= |\{xy \in E(G) : d_G(v, x) = d_G(v, y) \text{ and this common value is even}\}|, \end{aligned}$$

the WOWII invariants **dist_even**(v) and **even horizontal**(v). The convention $d_G(v, v) = 0$ makes v itself contribute to $\text{dist}_{\text{even}}(v)$; DeLaViña’s page examples imply this convention, and we verify below (Remark 4.3) that our conclusions do not depend on it. Finally, $n \bmod \Delta$ is an *atomic* invariant of the WOWII vocabulary (definition 32 of [9], defined when $\Delta \geq 2$), not a derived expression; this matters in Section 5.3.

Definition 2.1 (Clique blow-up). Let H be a graph with $V(H) = \{1, \dots, k\}$ and let $m \geq 1$. The *clique blow-up* $H[K_m]$ (the lexicographic product of H with K_m) has vertex set $V(H) \times \{1, \dots, m\}$, with (i, j) adjacent to (i', j') if and only if either $i = i'$ and $j \neq j'$, or $ii' \in E(H)$. We write $B_i = \{i\} \times \{1, \dots, m\}$ for the *blob* over i , and for $S \subseteq V(H[K_m])$ we write $s_i = |S \cap B_i|$ and call $s = (s_1, \dots, s_k)$ the *profile* of S .

Each B_i induces a K_m ; if $ii' \in E(H)$ then $B_i \cup B_{i'}$ induces a K_{2m} ; if $ii' \notin E(H)$ and $i \neq i'$ then no edge joins B_i to $B_{i'}$.

3 Pinning

Throughout this section H is a graph on $k \geq 1$ vertices, $m \geq 1$, and $G = H[K_m]$.

Lemma 3.1 (Blob profile lemma). *Let $S \subseteq V(G)$ be such that $G[S]$ is triangle-free, and let s be its profile. Put*

$$A = \{i \in V(H) : s_i = 2\}, \quad B = \{i \in V(H) : s_i = 1\}.$$

Then:

- (i) $s_i \leq 2$ for every i ;
- (ii) if $i \in A$ then $s_j = 0$ for every $j \in N_H(i)$; in particular A is independent in H and no edge of H joins A to B ;
- (iii) $G[S] \cong |A| \cdot K_2 \sqcup H[B]$ (a disjoint union of $|A|$ copies of K_2 with a copy of the subgraph of H induced by B).

Conversely, if $m \geq 2$ and (A, B) is any pair of disjoint subsets of $V(H)$ such that no edge of H joins a vertex of A to a vertex of $A \cup B$, then some $S \subseteq V(G)$ with $A = \{i : s_i = 2\}$, $B = \{i : s_i = 1\}$ exists and satisfies $G[S] \cong |A| \cdot K_2 \sqcup H[B]$.

Proof. (i) B_i induces a clique, so any three vertices of $S \cap B_i$ would induce a triangle.

(ii) Suppose $i \in A$, say $S \cap B_i = \{u, v\}$, and suppose $j \in N_H(i)$ has $s_j \geq 1$, say $w \in S \cap B_j$. Then $B_i \cup B_j$ induces a clique, so $\{u, v, w\}$ induces a triangle in $G[S]$, a contradiction. If $i, i' \in A$ were adjacent in H we would have $s_{i'} = 2 \geq 1$, contradicting what we just proved; the same computation excludes edges from A to B .

(iii) By (ii) the vertices of S lying in blobs indexed by A have no neighbours in S outside their own blob, and each such blob contributes exactly two vertices, adjacent to each other: $|A|$ components isomorphic to K_2 . Every blob indexed by B contributes exactly one vertex, and two such vertices, in blobs i, j with $i \neq j$, are adjacent in G if and only if $ij \in E(H)$. Hence the union of the B -blob vertices induces a copy of $H[B]$, and by (ii) again there are no edges between the two parts.

For the converse, choose for each $i \in A$ two vertices of B_i (possible since $m \geq 2$) and for each $i \in B$ one vertex of B_i . Parts (i)–(iii) of the computation above apply verbatim and give the stated isomorphism type, which is triangle-free whenever $H[B]$ is. $\square$

Theorem 3.2 (Pinning). *Let H be a graph and $m \geq 1$, $G = H[K_m]$. Then*

- (i) $\alpha(G) = \alpha(H)$;
- (ii) $\text{path}(G) \leq \text{tree}(G) \leq f(G) \leq b(G) \leq 2\alpha(H)$;
- (iii) $b(H) \leq b(G)$ and $\text{path}(H) \leq \text{path}(G)$, and the same for f and tree ;
- (iv) consequently, if $\text{path}(H) = 2\alpha(H)$ then $\text{path}(G) = \text{tree}(G) = f(G) = b(G) = 2\alpha(H)$ for every $m \geq 1$.

In particular each of these four quantities is independent of m under the hypothesis of (iv), while $n(G) = m \cdot n(H)$ and $\Delta(G) = m(1 + \Delta(H)) - 1$ grow linearly in m .

Proof. (i) An independent set of G meets each blob in at most one vertex (blobs are cliques) and its set of blob indices is independent in H (adjacent blobs are completely joined); hence $\alpha(G) \leq \alpha(H)$. Conversely one vertex per blob over a maximum independent set of H is independent in G .

(ii) The three left-hand inequalities are immediate from the inclusions $\{\text{paths}\} \subseteq \{\text{trees}\} \subseteq \{\text{forests}\} \subseteq \{\text{bipartite graphs}\}$, all four classes being closed under induced subgraphs. For the last, let S realise $b(G)$, so $G[S]$ is bipartite with parts X, Y . Both X and Y are independent in G , so $|X|, |Y| \leq \alpha(G) = \alpha(H)$ by (i), whence $b(G) = |X| + |Y| \leq 2\alpha(H)$.

(iii) If $H[S_0]$ is bipartite (resp. a forest, tree, path) for $S_0 \subseteq V(H)$, pick one vertex from B_i for each $i \in S_0$; the induced subgraph of G on the chosen vertices is isomorphic to $H[S_0]$.

(iv) By (iii), $\text{path}(G) \geq \text{path}(H) = 2\alpha(H)$, and by (ii) all four quantities lie between $\text{path}(G)$ and $2\alpha(H)$. $\square$

Remark 3.3. The inequality $b(G) \leq 2\alpha(G)$ used in the proof is elementary and holds for every graph. The content of Theorem 3.2 is that composing it with the exact identity $\alpha(H[K_m]) = \alpha(H)$ makes the bound *independent of m* , and that condition (iv) identifies when it is met with equality by all four numbers at once.

Corollary 3.4 (Odd cycles). *Let $r \geq 1$, $k = 2r + 1$, and $m \geq 1$. Then*

$$\alpha(C_k[K_m]) = r \quad \text{and} \quad \text{path}(C_k[K_m]) = \text{tree}(C_k[K_m]) = f(C_k[K_m]) = b(C_k[K_m]) = 2r.$$

Proof. $\alpha(C_{2r+1}) = r$, and deleting one vertex of C_{2r+1} leaves an induced path on $2r$ vertices, so $\text{path}(C_{2r+1}) = 2r = 2\alpha(C_{2r+1})$. Apply Theorem 3.2(i) and (iv). $\square$

For $r = 1$, $k = 5$ this is the pinning statement we need: for every $m \geq 1$,

$$\alpha(C_5[K_m]) = 2, \quad \text{path} = \text{tree} = f = b = 4 \quad \text{on } C_5[K_m]. \quad (1)$$

An explicit witness is worth recording, since it is the same set for all four numbers: taking one vertex from each of B_0, B_1, B_3, B_4 gives the induced path $B_1 - B_0 - B_4 - B_3$ on four vertices (the two chosen vertices in B_1 and B_3 are non-adjacent because 1 and 3 are at distance 2 on C_5).

Lemma 3.1 gives more than Theorem 3.2: it gives the exact value for an arbitrary hereditary class and shows it stabilises.

Proposition 3.5 (Exact values and stabilisation). *Let $\mathcal{F}$ be a class of graphs closed under isomorphism and induced subgraphs, all of whose members are triangle-free, and let $\mathcal{F}(G) = \max\{|S| : G[S] \in \mathcal{F}\}$. Then for every $m \geq 2$,*

$$\mathcal{F}(H[K_m]) = \max\{2|A| + |B|\},$$

the maximum being over all pairs of disjoint sets $A, B \subseteq V(H)$ such that no edge of H joins A to $A \cup B$ and $|A|K_2 \sqcup H[B] \in \mathcal{F}$; a quantity that does not depend on m . If moreover every member of $\mathcal{F}$ is bipartite, then $\mathcal{F}(H[K_m]) \leq 2\alpha(H)$ for every $m \geq 1$.

Proof. Members of $\mathcal{F}$ are triangle-free, so Lemma 3.1 applies to any S with $G[S] \in \mathcal{F}$, giving $|S| = 2|A| + |B|$ and $G[S] \cong |A|K_2 \sqcup H[B]$ for an admissible pair (A, B) ; this proves “ $\leq$ ”. The converse half of Lemma 3.1 proves “ $\geq$ ” when $m \geq 2$. For the last statement, let (A, B) be admissible with $|A|K_2 \sqcup H[B] \in \mathcal{F}$ bipartite. Then $H[B]$ is bipartite, so it has an independent set I with $|I| \geq |B|/2$. Since A is independent in H and has no H -neighbour in $B \supseteq I$, the set $A \cup I$ is independent in H , whence $\alpha(H) \geq |A| + |B|/2$, i.e. $2|A| + |B| \leq 2\alpha(H)$. $\square$

Example 3.6. For $H = C_5$ the admissible pairs achieving the maximum are $(\emptyset, V \setminus \{i\})$, giving P_4 ; $(\{i\}, \{i+2, i+3\})$, giving $K_2 \sqcup K_2$; and $(\{i, i+2\}, \emptyset)$, giving $2K_2$. All three have $2|A| + |B| = 4$. The first is a path, so all four numbers in (1) equal 4; the second and third are forests but not trees, which is why the value 4 is realised in three combinatorially different ways.

3.1 The growing side, and the cliff principle

Pinning is only half of the mechanism. The other half is that the invariants used as *correction terms* in the same conjectures are unbounded on the family.

Proposition 3.7 (Growth). *Let H be connected, non-complete, with $k = n(H) \geq 3$ vertices, and let $G = H[K_m]$. Then $n(G) = km$, G is $(m(1 + \Delta(H)) - 1)$ -regular when H is regular, $\omega(G) = m\omega(H)$, $\text{diam}(G) = \text{diam}(H)$, and for every $v = (i, a) \in V(G)$,*

$$\text{dist}_{\text{even}}(v) = 1 + m \cdot |\{j \in V(H) \setminus \{i\} : d_H(i, j) \text{ is even}\}|.$$

In particular each of n , Δ , ω , $\text{dist}_{\text{even}}$ is a linear, unbounded function of m whenever the corresponding coefficient is nonzero.

Proof. The order and regularity are immediate. A clique of G meets only blobs whose indices are pairwise equal or adjacent in H (non-adjacent blobs have no edges between them) and may contain a whole blob, so $\omega(G) = m\omega(H)$. Distinct vertices of the same blob are adjacent, and blobs over adjacent vertices of H are completely joined, so $d_G((i, a), (i', a')) = d_H(i, i')$ whenever $i \neq i'$, and $d_G((i, a), (i, a')) = 1$ for $a \neq a'$. Since $\text{diam}(H) \geq 2$ this gives $\text{diam}(G) = \text{diam}(H)$ and the stated formula for $\text{dist}_{\text{even}}$: the vertices at even distance from (i, a) are (i, a) itself (distance 0) together with the whole of B_j for each $j \neq i$ with $d_H(i, j)$ even. $\square$

The failure principle is now a one-line consequence, but stating it is the point of the paper.

Proposition 3.8 (Cliff principle). *Let $(G_m)_{m \geq 1}$ be a family of graphs and let P, Φ be graph invariants such that $P(G_m) = c$ for all m and $\Phi(G_m) \rightarrow \infty$. Then the statement “ $P(G) \geq \Phi(G)$ for all G in the relevant class” fails on G_m for every m with $\Phi(G_m) > c$. If $m \mapsto \Phi(G_m)$ is nondecreasing and m_0 is the largest index with $\Phi(G_{m_0}) \leq c$, then the statement holds on $G_1, \dots, G_{m_0}$ and fails on every G_m with $m > m_0$; if in addition $\Phi(G_{m_0}) = c$ then G_{m_0} satisfies the statement with equality. Symmetrically, if $\Phi(G_m) \rightarrow -\infty$ then “ $P(G) \leq \Phi(G)$ ” fails from some index onwards, again with equality at the last surviving index when $\Phi(G_{m_0}) = c$.*

The observation that matters for conjecture generation is the last clause. *The last surviving member is an equality case.* A generator whose database stops at G_{m_0} therefore does not merely fail to refute the candidate; it records the candidate as *sharp*, and a significance heuristic that rewards sharpness promotes it. The database boundary and the sharpness signal coincide. We return to this in Section 6.

4 The family $C_5[K_m]$

We now specialise to $H = C_5$ and record the exact values of everything used later. Index the blobs by $\mathbb{Z}_5$, so $B_i \cup B_{i+1}$ induces K_{2m} and there are no edges between B_i and B_{i+2} .

Proposition 4.1 (Basic invariants). *Let $m \geq 1$ and $G = C_5[K_m]$. Then*

$$n(G) = 5m, \quad |E(G)| = \frac{5m(3m-1)}{2}, \quad \text{diam}(G) = \text{rad}(G) = 2,$$

G is $(3m-1)$ -regular and vertex-transitive, and

$$\alpha(G) = 2, \quad \omega(G) = 2m, \quad \gamma(G) = 2, \quad \gamma_t(G) = 3, \quad \gamma_c(G) = 3, \quad L_s(G) = 5m - 3,$$

and the local independence number $\lambda(v) := \alpha(G[N(v)])$ equals 2 at every vertex.

Proof. Order, size, regularity and vertex-transitivity are immediate from the construction ($\deg = (m - 1) + 2m$; the automorphism group contains the dihedral action on blobs and all within-blob permutations, acting transitively). Non-adjacent vertices lie in blobs at C_5 -distance 2 and have a common neighbour in an intermediate blob, so $\text{diam}(G) = 2$; every vertex has non-neighbours, so $\text{rad}(G) = 2$. $\alpha(G) = \alpha(C_5) = 2$ and $\omega(G) = 2m$ by Theorem 3.2(i) and Proposition 3.7.

For domination: a single vertex of B_i dominates exactly $B_{i-1} \cup B_i \cup B_{i+1}$, so $\gamma(G) \geq 2$, and $\{u, w\}$ with $u \in B_i$, $w \in B_{i+2}$ dominates all five blobs, so $\gamma(G) = 2$. Since a total dominating set of size 2 must be an edge uv with $N(u) \cup N(v) = V(G)$, and the union of two closed blob-neighbourhoods $\{i-1, i, i+1\} \cup \{i', i'-1, i'+1\}$ with $i' \in \{i, i \pm 1\}$ never covers $\mathbb{Z}_5$, we get $\gamma_t(G) \geq 3$; and $\{u, v, w\}$ with $u \in B_0$, $v \in B_1$, $w \in B_2$ is total dominating (their blob neighbourhoods cover $\mathbb{Z}_5$ and each of the three has one of the others as a neighbour), so $\gamma_t(G) = 3$. A connected dominating set of size 2 is in particular a dominating set inducing an edge, so its blobs are equal or adjacent and again fail to cover $\mathbb{Z}_5$; the set $\{u, v, w\}$ above is connected and dominating, so $\gamma_c(G) = 3$ and $L_s(G) = n - \gamma_c = 5m - 3$.

Finally, for $v \in B_i$ we have $N(v) = (B_i \setminus \{v\}) \cup B_{i-1} \cup B_{i+1}$. An independent set of $G[N(v)]$ contains at most one vertex of each blob and its set of blob indices is independent in the path $(i-1, i, i+1)$ induced by C_5 on those three indices; hence $\lambda(v) \leq 2$. Taking one vertex of B_{i-1} and one of B_{i+1} , which are non-adjacent because $i-1$ and $i+1$ are at C_5 -distance 2, gives $\lambda(v) = 2$, for every $m \geq 1$. $\square$

Proposition 4.2 (Parity, residue and complement). *Let $m \geq 1$, $G = C_5[K_m]$, and $v \in V(G)$. Then*

$$\text{dist}_{\text{even}}(v) = 2m + 1, \quad \text{hor}_{\text{even}}(v) = m(2m - 1), \quad \overline{G} \cong C_5[\overline{K_m}],$$

$\overline{G}$ is $2m$ -regular, triangle-free, of diameter 2, and $\min_{e \in E(\overline{G})} |N_{\overline{G}}(e)| = 4m$, where $N_{\overline{G}}(xy) = N_{\overline{G}}(x) \cup N_{\overline{G}}(y)$. Moreover

$$n \bmod \Delta = 5m \bmod (3m - 1) = 2m + 1 \quad \text{for every } m \geq 3,$$

while $5 \bmod 2 = 1$ for $m = 1$ and $10 \bmod 5 = 0$ for $m = 2$.

Proof. Let $v \in B_i$. Vertices at distance 0 or 2 from v are v itself and all of $B_{i+2} \cup B_{i-2}$, giving $\text{dist}_{\text{even}}(v) = 1 + 2m$; the remaining $3m - 1$ vertices are at distance 1. The vertex pairs at equal even distance from v are the pairs inside $B_{i+2} \cup B_{i-2}$ (both at distance 2); since $i+2$ and $i-2 = i+3$ are adjacent on C_5 , this set induces K_{2m} , contributing $\binom{2m}{2} = m(2m - 1)$ edges, and v has no neighbour at distance 2 from itself, so $\text{hor}_{\text{even}}(v) = m(2m - 1)$. (The same value is obtained under the weaker reading “both endpoints at even distance”, because v is isolated from $B_{i+2} \cup B_{i-2}$.)

In $\overline{G}$, (i, j) and (i', j') are adjacent exactly when $i - i' \equiv \pm 2 \pmod{5}$; since $\{\pm 2\}$ generates a 5-cycle on $\mathbb{Z}_5$, $\overline{G}$ is the blow-up of a 5-cycle by independent sets, so it is $2m$ -regular, triangle-free (an odd cycle blow-up by independent sets contains a triangle only if the base does) and of diameter 2. Let $xy \in E(\overline{G})$ with $x \in B_i$, $y \in B_{i+2}$. Then $N_{\overline{G}}(x) = B_{i+2} \cup B_{i-2}$ and $N_{\overline{G}}(y) = B_{i+4} \cup B_i$; these four blobs are distinct, so $|N_{\overline{G}}(x) \cup N_{\overline{G}}(y)| = 4m$, independently of the edge chosen.

Finally $5m = 1 \cdot (3m - 1) + (2m + 1)$ and $0 \leq 2m + 1 < 3m - 1$ exactly when $m > 2$. $\square$

Remark 4.3. Two conventions in the WOWII vocabulary are ambiguous in the published text: whether v counts in $\text{dist}_{\text{even}}(v)$, and whether the endpoints of e count in $N(e)$. Under the alternative conventions, $\text{dist}_{\text{even}}(v) = 2m$ and $|N_{\overline{G}}(e)| = 4m - 2$. All four refutations in Section 5 survive all combinations of these conventions; the per-conjecture verdicts for every combination are recorded in the sweep data (Section 10).

Combining Proposition 4.1, Proposition 4.2 and (1), the family exhibits the mechanism in its purest form:

pinned (constant in m)	growing (linear or quadratic in m)
$\alpha = 2$	$n = 5m, \quad \Delta = \delta = 3m - 1, \quad \omega = 2m$
$\text{path} = \text{tree} = f = b = 4$	$\text{dist}_{\text{even}} = 2m + 1, \quad n \bmod \Delta = 2m + 1 \ (m \geq 3)$
$\gamma = 2, \quad \gamma_t = 3, \quad \gamma_c = 3$	$\text{hor}_{\text{even}} = m(2m - 1), \quad \min N_{\overline{G}}(e) = 4m$
$\text{diam} = \text{rad} = 2, \quad \lambda(v) = 2$	$L_s = 5m - 3, \quad \deg_{\text{avg}}(\overline{G}) = 2m$

5 Four conjectures, four cliffs

We now state the four WOWII conjectures verbatim as transcribed from DeLaViña's page [8] and evaluate them on the family. The transcriptions, including the unbalanced brackets in 64, reproduce the page as published; the transcription audit and the recovery of DeLaViña's own invariant-definition database are described in Section 7.

5.1 Conjecture 63

WOWII 63 (page marker 0, dated Mar. 25, 2004). *If G is a simple connected graph, then $f(G) \geq \lceil (\min_v \text{dist}_{\text{even}}(v) + b(G) + 1)/3 \rceil$.*

Theorem 5.1. *On $G = C_5[K_m]$ the right-hand side of WOWII 63 equals $\lceil (2m+6)/3 \rceil$. Hence Conjecture 63 holds on $C_5[K_m]$ for $m \leq 3$, with equality for $m \in \{2, 3\}$, and fails for every $m \geq 4$.*

Proof. By Proposition 4.2 $\min_v \text{dist}_{\text{even}}(v) = 2m + 1$, and by (1) $b = 4$ and $f = 4$. So the right-hand side is $\lceil (2m + 1 + 4 + 1)/3 \rceil = \lceil (2m + 6)/3 \rceil$, which is ≤ 4 exactly when $2m + 6 \leq 12$, i.e. $m \leq 3$, and equals 4 exactly when $10 \leq 2m + 6 \leq 12$, i.e. $m \in \{2, 3\}$. $\square$

5.2 Conjecture 85

WOWII 85 (page marker 0, dated Apr. 04, 2004). *If G is a simple connected graph, then $\text{tree}(G) \geq \lceil \sqrt{1 + 2 \min_v \text{dist}_{\text{even}}(v)} \rceil$.*

Theorem 5.2. *On $G = C_5[K_m]$ the right-hand side of WOWII 85 equals $\lceil \sqrt{4m + 3} \rceil$. Hence Conjecture 85 holds on $C_5[K_m]$ for $m \leq 3$, with equality for $m \in \{2, 3\}$, and fails for every $m \geq 4$.*

Proof. $1 + 2(2m + 1) = 4m + 3$, and $\text{tree} = 4$. Now $\lceil \sqrt{4m + 3} \rceil \leq 4$ iff $4m + 3 \leq 16$ iff $m \leq 3$ (as m is an integer), and $\lceil \sqrt{4m + 3} \rceil = 4$ iff $9 < 4m + 3 \leq 16$ iff $m \in \{2, 3\}$. $\square$

5.3 Conjecture 64

WOWII 64 (page marker 0, dated Mar. 25, 2004). *If G is a simple connected graph, then $f(G) \geq \text{CEIL}[\sqrt{\alpha(G) \cdot (1 + (n \bmod \Delta(G)))}]$.*

The bracketing on the page is unbalanced. The reading we use is

$$f(G) \geq \left\lceil \sqrt{\alpha(G) (1 + (n \bmod \Delta(G)))} \right\rceil,$$

which is pinned by the page's own `printDefinitions` link with arguments (41, 5, 32, 16, 0): the constituents are declared to be the forest number, α , the *atomic* invariant “ n modulus maximum degree” (definition 32), and the ceiling function.

Theorem 5.3. *On $G = C_5[K_m]$ with $m \geq 3$, the right-hand side of WOWII 64 equals $\lceil 2\sqrt{m+1} \rceil$; for $m \in \{1, 2\}$ it equals 2. Hence Conjecture 64 holds on $C_5[K_m]$ for $m \leq 3$, with equality at $m = 3$, and fails for every $m \geq 4$.*

Proof. For $m \geq 3$, Proposition 4.2 gives $n \bmod \Delta = 2m + 1$, and $\alpha = 2$, so the radicand is $2(2m + 2) = 4(m + 1)$ and the right-hand side is $\lceil 2\sqrt{m+1} \rceil$. This is ≤ 4 iff $4(m + 1) \leq 16$ iff $m \leq 3$, with equality exactly at $m = 3$ (where $4(m + 1) = 16$ is a perfect square). For $m = 1$ the radicand is $2(1 + 1) = 4$ and for $m = 2$ it is $2(1 + 0) = 2$, giving right-hand sides 2 and 2. Since $f = 4$ throughout, the claim follows. $\square$

5.4 Conjecture 309

WOWII 309 (page marker 0, dated Mar. 1, 2007). *If G is a simple connected graph such that $n(G) > 2$, then*

$$\gamma_t(G) \leq \frac{1}{2} \left[\max_{v \in V(G)} \{ \text{dist}_{\text{even}}(v) - \text{hor}_{\text{even}}(v) \} + \min_{e \in E(G)} |N_{\overline{G}}(e)| \right].$$

m	n	Δ	pinned	min dist _{even}	RHS ₆₃	RHS ₈₅	RHS ₆₄	RHS ₃₀₉
1	5	2	4	3	3	3	2	3
2	10	5	4	5	4	4	2	7/2
3	15	8	4	7	4	4	4	2
4	20	11	4	9	5	5	5	-3/2
5	25	14	4	11	6	5	5	-7

Table 1: The discretization cliff on $C_5[K_m]$. The column “pinned” is the common value $f = b = \text{tree} = \text{path} = 4$; the left-hand side of 63 and 64 is f , of 85 is tree , and of 309 is $\gamma_t = 3$. A conjecture holds at m iff its RHS is at most its LHS. Bold entries are exact equalities. At $m = 1, 2$ all four hold, with 309 tight at $m = 1$ and 63, 85 tight at $m = 2$; at $m = 3$ conjectures 63, 85, 64 are simultaneously tight while 309 already fails; for every $m \geq 4$ all four fail. Columns 63, 85, 64 are printed by `scripts/family_sweep.py`; column 309 is Theorem 5.4. All entries are re-derived by `verify_paper_claims.py` (Section 10).

Theorem 5.4. *On $G = C_5[K_m]$ the right-hand side of WOWII 309 equals $\frac{1}{2}(-2m^2 + 7m + 1)$. Hence Conjecture 309 holds on $C_5[K_m]$ for $m \in \{1, 2\}$, with equality at $m = 1$, and fails for every $m \geq 3$, with margin $\frac{1}{2}(2m - 5)(m - 1) \rightarrow \infty$.*

Proof. By Proposition 4.2 and vertex-transitivity, $\text{dist}_{\text{even}}(v) - \text{hor}_{\text{even}}(v) = 2m + 1 - m(2m - 1) = -2m^2 + 3m + 1$ for every v , and $\min_e |N_G(e)| = 4m$. So the right-hand side is $\frac{1}{2}(-2m^2 + 3m + 1 + 4m) = \frac{1}{2}(-2m^2 + 7m + 1)$. With $\gamma_t = 3$ (Proposition 4.1), the conjecture asserts $6 \leq -2m^2 + 7m + 1$, i.e. $2m^2 - 7m + 5 \leq 0$, i.e. $(2m - 5)(m - 1) \leq 0$, which for positive integers m holds exactly for $m \in \{1, 2\}$, with equality only at $m = 1$. $\square$

Theorem 5.4 recovers Gebendorfer’s closed forms [14] by an independent derivation; the smallest member of the family that refutes 309 is $C_5[K_3]$, of order 15, where the conjecture asserts $3 \leq 2$.

6 The cliff table and the $m = 3$ diagnostic

Table 1 collects the four theorems of Section 5.

Three features of the table are worth isolating.

(a) Simultaneity. Three conjectures from two different sections of the list (63 and 64 sit in “2nd run on lower bounds on the forest number”; 85 in “lower bounds on the tree number”) have their last surviving member at the *same* index $m = 3$, and all three are attained there with equality. They are not the same statement, and they are not rescaled versions of one another: their right-hand sides grow like $\frac{2}{3}m$, $2\sqrt{m}$ and $2\sqrt{m}$ respectively, and are built from different invariants — $\text{dist}_{\text{even}}$ alone; $\text{dist}_{\text{even}}$ under a square root; and α together with the residue $n \bmod \Delta$. The synchronisation comes from the pinning: all three lower-bound a quantity stuck at 4, so three unbounded functions that are still at most 4 when $m = 3$ have all passed it by $m = 4$. Simultaneity is therefore what the mechanism predicts, and it is the reason a single family can clear several entries of a corpus at once rather than one.

(b) The equalities are partly manufactured by the rounding. At $m = 3$ the three right-hand sides are $(2 \cdot 3 + 6)/3 = 4$ exactly (the numerator is divisible by 3), $\lceil \sqrt{15} \rceil = 4$ because 15 falls in the gap $(9, 16]$, and $\sqrt{2 \cdot 8} = 4$ exactly because 16 is a perfect square. Two of these equalities are genuine, in the sense that they survive deleting the ceiling; the third (85 at $m = 3$) exists only because of it, as do both equalities at $m = 2$. Of the five bold entries in Table 1 for 63, 85 and 64, three are artefacts of rounding and two are not.

Deleting the ceilings does *not* move the failure thresholds: $4 \geq (2m + 6)/3$, $4 \geq \sqrt{4m + 3}$ and $4 \geq 2\sqrt{m + 1}$ each hold exactly for integers $m \leq 3$. So the rounding does not create the cliff — the pinning and the growth do that — but it creates three of the five equalities *at the edge of the cliff*, and the equality is precisely what a Dalmatian-style significance heuristic reads. This is the sense in which the phenomenon is a *discretization* cliff: the discretization converts a bound with small slack, on the last few members before the failure, into a bound that looks sharp.

(c) **The last surviving member is an extremal witness.** This is Proposition 3.8’s last clause instantiated. A database containing $C_5[K_1]$, $C_5[K_2]$ and $C_5[K_3]$ but not $C_5[K_4]$ would report all three of 63, 85, 64 as true and sharp. That is the strongest possible endorsement a Graffiti-type filter can give.

We therefore propose the following diagnostic, and we state it as a heuristic rather than a theorem.

Diagnostic. Let $\mathcal{G}$ be a family of graphs indexed so that some invariant is pinned along it. If a machine-generated bound is attained with equality on the largest member of $\mathcal{G}$ present in the generator’s database, and if its right-hand side is a nonconstant function of the index, then the bound is a candidate for refutation by the next member.

The diagnostic has real content only if the two hypotheses can be checked without knowing the database. The first can: one computes the slack of the bound on successive members of a candidate family and looks for a zero. The second can: one checks whether the invariants occurring on the right-hand side are among those that grow. Both are cheap. What we do *not* claim is that following the diagnostic locates new counterexamples at a useful rate; see Section 8.

Remark 6.1 (On the database boundary). It is tempting to say that Graffiti.pc’s verification database stopped at some order n_0 with $15 \leq n_0 < 20$. We have no evidence for any particular n_0 : the exact contents of the database used for the 2004–2007 WOWII runs are not published, as far as we can determine [CITATION NEEDED]. What is verifiable is a lower bound on where the failure begins: each of 63, 85 and 64 holds on all 995 connected graphs of order at most 7 and on a list of standard larger graphs, and 309 holds on all 989 applicable connected graphs of order 3 to 7 with 36 exact equalities among them (Section 10). Neither do we claim that $C_5[K_3]$ itself was in the database; the equality at $m = 3$ is a fact about the family, and its coincidence with a plausible database horizon is a hypothesis about why the conjectures were emitted.

7 The family against the whole WOWII corpus

7.1 Protocol

The WOWII page was transcribed into 522 machine-readable records (521 distinct identifiers; one entry, 434a, occurs twice on the page). Each record carries the verbatim statement, the page’s status marker, and its section. The corpus splits as 220 open, 139 already refuted by others, and 163 proved or otherwise resolved.

Evaluating a WOWII statement on a concrete graph is not mechanical, for two reasons. First, the page was authored in a Symbol-font HTML dialect and several statements are ambiguous in precedence or in the scope of a set-valued argument. Second, some statements are simply corrupt as published. We handled both as follows.

1. **Reading enumeration.** For each ambiguous statement we enumerated all plausible readings and evaluated every one; verdicts are reported per reading.
2. **Database-sanity gate.** A reading is discarded if it is violated by a graph that was certainly available to the generator — concretely, by one of the 995 connected graphs of order $2 \leq n \leq 7$, or by a standard small named graph. A conjecture the program emitted cannot be false on the program’s own trivial range; a reading that is false there is a mis-transcription, not a discovery. This gate discarded readings for 10 statements and disqualified 3 statements entirely (below).
3. **Invariant definitions.** All invariants were implemented against DeLaViña’s own definition database `wowIIdefs.js`, recovered from an archived snapshot of the site, so that “ $\text{dist}_{\text{even}}$ ”, “ hor_{even} ”, “ $N(e)$ ”, “ L_s ”, “ maxine ” and the rest carry the author’s definitions rather than ours.
4. **Adversarial re-verification.** Every VIOLATED verdict was recomputed by a second, independently written code path whose stated objective was to *save* the conjecture. Two of the six initial open-list violations did not survive this step (see 7.3).

7.2 Results

Against $C_5[K_4]$ the outcome is:

#	statement (as transcribed)	pinned LHS	growing term	LHS	RHS
24	$b \geq \lambda + \lceil \min \text{dist}_{\text{even}}/3 \rceil$	$b = 4$	$\text{dist}_{\text{even}} = 2m + 1$	4	5
25	$b \geq 2\lceil (1 + \min \text{dist}_{\text{even}})/3 \rceil$	$b = 4$	$\text{dist}_{\text{even}} = 2m + 1$	4	8
46	$f \geq \lfloor \text{path} - 1 + \frac{1}{3}(n \bmod \Delta) \rfloor$	$f = 4$	$n \bmod \Delta = 2m + 1$	4	6
49	$f \geq \lceil 2 + \frac{1}{6} \text{length}(\overline{G}) \rceil$	$f = 4$	$\text{length}(\overline{G}) = 2m\sqrt{5m}$	4	8
52	$f \geq \lceil \frac{1}{2}(dd + 1 + (n \bmod \Delta)) \rceil$	$f = 4$	$n \bmod \Delta = 2m + 1$	4	6
54	$f \geq \lceil \text{dist}_{\text{avg}}(V) + \frac{1}{2} \min \text{dist}_{\text{even}} \rceil$	$f = 4$	$\text{dist}_{\text{even}} = 2m + 1$	4	6
55	$f \geq \lceil \min \text{dist}_{\text{even}} - 1 + N(A) /3 \rceil$	$f = 4$	$\text{dist}_{\text{even}}, N(A) = 5m$	4	15
56	$f \geq \lceil \sqrt{\text{dist}_{\text{max}}(A)(1 + \deg_{\text{avg}}(\overline{G}))} \rceil$	$f = 4$	$\deg_{\text{avg}}(\overline{G}) = 2m$	4	5

Table 2: The eight retro-kills on $C_5[K_4]$. In every case a pinned induced-subgraph number is lower-bounded by a term that Proposition 3.7 makes unbounded in m . Here $\lambda = \max_v \alpha(G[N(v)]) = 2$, dd is the number of distinct degrees ($= 1$, the graph is regular), A is the set of minimum-degree vertices ($= V$), and $\text{length}(G) = \sqrt{\sum_v \deg(v)^2}$. LHS/RHS columns are the values at $m = 4$ from **results/refuted_sweep/**.

subcorpus	records	holds	n/a	violated
open	220	167	47	6
refuted	139	101	29	8 (+1)
proved	163	98+19	45	1
total	522			

The four open kills. 63, 85, 64, 309; Section 5. Conjectures 63 and 85 were first refuted by $C_5[K_4]$ in July 2026; 64 and 309 were subsequently refuted by Gebendorfer using the same carrier family, with the earlier certificate credited in both notes [13, 14]. Formal Lean 4 certificates without **sorry** exist for all four.

Eight retro-kills. Among the 139 conjectures already refuted by others, $C_5[K_4]$ is a counterexample to 24, 25, 46, 49, 52, 54, 55, 56 (and, under one of two readings, 77). These are not novelty claims — the conjectures were dead already. They are evidence about the mechanism, because they are exactly the predicted shape. Table 2 lists them with the pinned left-hand side and the growing right-hand term.

The sharpness gallery. This is the larger half of the data and, we argue, the more informative half. Against the corpus, $C_5[K_4]$ attains zero slack in:

- **19 proved theorems** of the corpus: 4, 7, 15, 16, 18, 37, 57, 68, 89, 94, 99, 173, 382a, 409a, 411, 420a, 451, 452, 458. For instance $b = 4 = 2\text{rad}(G) = 2\alpha(G)$ and $\alpha(G) = 2 = b(G) - \min_v \lambda(v)$ hold simultaneously, so several of these theorems are extremal on the same graph at once.
- **12 conjectures open at the time of the sweep**, under gate-surviving readings: 19, 174, 176, 181, 182, 183, 184, 185, 382e, 422b, 430a, 438b. (Three of these have since moved: 174 was refuted by a third party in July 2026 by an 11-vertex graph; 438b was subsequently proved; 176, 181 and 430a were later refuted, by other graphs, in work not reported here.)
- **Four already-refuted bounds**: 219, 289, 424, 430b.
- One of the three conjectures (137) that the July 2026 community transcription of the page omitted altogether and that our audit recovered: $\text{path} = 4 = 4/p(\overline{G})$, the complement being Hamiltonian.

That is roughly three dozen zero-slack coincidences for one graph. Some of them have a common explanation and it is instructive. WOWII 173 is a theorem of DeLaViña and Waller [10]: $L_s(G) + b(G) \geq n(G) + 1$ for every nontrivial connected G . If $\text{diam}(G) = 2$ then $G^2 = K_n$, and under the gate-surviving readings the right-hand sides of 176, 182, 183, 184 and 185 all collapse to exactly $n + 1$. Our family has $L_s + b = (5m - 3) + 4 = 5m + 1 = n + 1$: it lies on the sharpness locus of a known theorem, and the five later conjectures are machine-generated attempted strengthenings whose correction terms vanish there. No diameter-two graph can refute any of them. This is a concrete instance of a general and, we think, under-appreciated point: a large block of zero-slack results on one carrier can be a signal that several published conjectures share a single already-proved baseline, which is diagnostic information about the corpus rather than about the graph.

7.3 Corpus defects, stated rather than exploited

Four entries of the corpus are corrupt as published, and we do not count them.

- 412f and 448b are violated by $C_5[K_4]$ under the literal published wording — and also by K_3 and by C_4 respectively, i.e. by graphs certainly inside the generator’s verification range. Under every repair of the wording that is clean on the 995-graph gate, $C_5[K_4]$ satisfies them. As published, 412f is violated by 62% of all connected graphs of order at most 7. A “refutation” of a page typo is empty, so these are recorded as NOT A USABLE KILL. We checked archived captures back to 2010 (for 412f) and 2016 (for 448b) and found the same corrupt wording already present, so the corruption is not recent.
- 401b, on which $C_5[K_4]$ is exactly tight, has a transcription that is violated by $K_{1,5}$ and $K_{1,6}$; its tightness is therefore not interpretable and it is excluded from the sharpness count above.
- **Theorem 97** of the corpus, as transcribed, reads $\alpha(G) \leq \max_v \lambda(v) - \delta(\overline{G})$ and is falsified by $C_5[K_4]$ — and by C_5 and by the Petersen graph. The published minus sign is almost certainly a plus lost in the Symbol-font encoding; with a plus the statement is true and holds here with slack 8. This is the single ANOMALY_VIOLATED entry in the proved subcorpus and it is a transcription artefact, not a counterexample to a theorem.

We also record, as a caveat on any exhaustive claim about this corpus, that the July 2026 community parse of the page missed three open conjectures entirely (136, 137, 138; their identifier cells are split across HTML tags in a way that defeats a naive parser) and that the page itself was edited on 2026-08-06, moving ten conjectures from open to resolved. Any statement of the form “all N open WOWII conjectures” is a statement about a snapshot.

7.4 Where the carrier is useless

The other side of an exhaustive sweep is knowing the boundary of a tool’s reach. On $C_5[K_4]$: the 47 “not applicable” open records are almost all conjectures with tree, bipartite, triangle-free, planar or sparse hypotheses, which the graph fails; the well-total-dominated characterisations 314–328 all escape vacuously (the graph is not well-total-dominated, but every antecedent fails on it, with 314 saved only by the graph’s 260 triangles); and the remaining γ_t upper bounds 305, 308, 310 hold with slack 5 or more. The carrier is a specialised instrument: it is productive precisely against machine-generated inequalities that combine a pinned induced-subgraph or domination invariant with a density/parity correction, and it is inert everywhere else. A separate sweep of 71 AutoGraphiX targets produced no gate-surviving violation (45 hold, 24 have unrecoverable definitions, one is inapplicable and already proved, and the single apparent violation fails the database-sanity gate), and a sweep of a set of human structural conjectures produced none either.

8 Limitations, and what this is not

This is not a counterexample engine. We have exhibited one family, analysed one base graph $H = C_5$ in detail, and refuted four open conjectures in one corpus. The pinning theorem is general and cheap to instantiate for other H (odd cycles all work, by Corollary 3.4), but the *other* half of the mechanism — the requirement that the target corpus contains bounds combining a pinned invariant with a growing one — is a property of the corpus, not of the construction. We have no evidence that this regime is common outside machine-generated inequality corpora. Applied to human structural conjectures the same carrier produced nothing.

We do not claim prospective navigation. A natural conjecture suggested by Section 6 is that the equality data can be used *forward*: identify the invariant that keeps a family pinned to an equality wall, construct a family that changes that invariant while preserving the rest of the extremal structure, and cross. We have one suggestive instance and we present it as exactly that, with no efficacy claim attached.

Observation 8.1 (Suggestive follow-up; not a result of this paper). The whole blow-up family saturates the “ $L_s + b$ ” cluster of WOWII (conjectures 174–186) at exact equality without ever crossing it, and every member has $\alpha = \lambda_{\max}$. WOWII 181, $L_s + b \geq \alpha + \deg_{\text{avg}}(B(G^2))$, requires $\alpha > \lambda_{\max}$ to be violated. The triangular graph $T(7) = L(K_7)$, the strongly regular $(21, 10, 5, 4)$ graph, has $\alpha = 3 > 2 = \lambda_{\max}$,

diameter 2, $b = 6$ and $\gamma_c = 5$ so $L_s = 16$; hence $L_s + b = 22 < 23 = 3 + 20$, and $T(n)$ for $n \geq 7$ gives an infinite family with margin $\rightarrow \infty$. This violation is *reading-dependent*: under the alternative parse that measures $\deg_{\text{avg}}$ in G rather than in G^2 , the conjecture holds with slack 9. We report it because it is the cleanest instance of the pattern we have, not as evidence that the pattern is a method.

Against that single instance stands the following. In our own broader experimentation the tightness-navigation idea has been applied under a frozen, design-before-evaluation protocol many dozens of times, in and out of graph theory, and has produced essentially no crossings that a plain catalogue search did not also find at comparable cost. The competing hypothesis — that machine conjecture corpora are simply under-verified enough that any systematic search refutes at this rate, and that tightness contributes narrative rather than guidance — is consistent with everything we have observed, including the fact that an independent researcher refuted two further conjectures with the same carrier within days of its publication. A proper test would preregister three arms (catalogue, generic random search, wall navigation) against a freshly generated conjecture population with equal budget, and report unique crossings. We have not run that test and therefore make no claim.

We do not claim knowledge of the historical database. See the remark at the end of Section 6. The mechanism is proved; the historical explanation of why these particular conjectures were emitted is a hypothesis consistent with the equality data.

Transcription risk is real and not fully eliminable. Every claim above is a claim about a reading of a statement rendered in a lossy encoding. We mitigated by enumerating readings, by gating readings against the generator’s own trivial range, and by implementing the author’s own definitions; we did not eliminate the risk. Four corpus entries are demonstrably corrupt (§7.3); there may be more.

9 Related work

The Graffiti lineage. Fajtlowicz’s *Graffiti* [11] introduced the design: a finite database of graphs with precomputed invariants, a generator of candidate inequalities, and a significance filter. DeLaViña’s *Graffiti.pc* [6], with historical context in [7], is the variant that produced the WOWII list [8] analysed here. Larson’s survey [16] in the same volume sets out the general problem of automated conjecture-making. Larson and Van Cleemput [17] reconstructed and formalised Fajtlowicz’s Dalmatian heuristic and released the *CONJECTURING* program; their analysis of the sharpness criterion is the closest prior discussion we know of to the mechanism studied here, though it is concerned with the heuristic’s completeness properties rather than with its failure modes on structured families. Davila’s *TxGraffiti* [4] and its agentic successor *The Optimist* [3] continue the lineage with linear-programming-based generation; [5] reports ten years of open conjectures from that system. The AutoGraphiX system of Caporossi and Hansen [2] takes a different route — variable neighbourhood search over graph space rather than filtering against a fixed database — and the survey of Aouchiche and Hansen [1] documents its output in spectral graph theory. The distinction matters for our thesis: a search-based generator explores the graph space in the direction of extremality and so is less exposed to the specific blind spot we describe, whereas a database-filtered generator’s blind spot is exactly the complement of its database.

Formal corpora. The `google-deepmind/formal-conjectures` project [12] is a growing Lean 4 formalisation of open conjectures, including a substantial batch of WOWII statements. It changes the practical situation in two ways. First, it pins readings: a formalised statement has one meaning, and the ambiguity discussed in Section 7 is resolved (though possibly resolved *differently* from the author’s intent, which is itself a finding worth reporting upstream). Second, it makes refutations checkable: a counterexample can be shipped as a **sorry-free** Lean term rather than as a table of numbers. All four refutations of Section 5 carry such certificates. We note, without developing the point here, that the same finite-database blind spot applies to *benchmark* curation: a conjecture corpus assembled to test automated provers inherits whatever verification horizon its sources had, and statements that are false just past that horizon are a systematic contaminant of such benchmarks.

Lexicographic products. The identity $\alpha(G[H]) = \alpha(G)\alpha(H)$ used in Theorem 3.2(i) is classical; the independence number of a lexicographic product is one of the functions treated by Geller and Stahl [15], whose paper is better known for the corresponding statement about the chromatic number. Clique

blow-ups of C_5 appear throughout extremal graph theory as sharpness examples — most familiarly for Ramsey-type and Reed-type bounds — but we are not aware of a prior treatment of them as diagnostic objects for machine-generated conjecture corpora.

10 Reproducibility

Every numerical claim above is reproducible from the public repository <https://github.com/Kuberwastaken/c5-k4>. Concretely:

- `scripts/family_sweep.py` prints Table 1’s columns for 63, 85, 64 over $m = 1, \dots, 5$ by exhaustive subset enumeration (no solver). Requires only `networkx`.
- `scripts/verify_wowii_64.py` is a self-contained exhaustive verifier for the $C_5[K_4]$ refutation of 64: it re-derives $\alpha = 2$ from all 1140 triples, $f = 4$ from all $\binom{20}{5} = 15,504$ five-subsets, and the arithmetic of the right-hand side.
- `scripts/verify_wowii_309.py` re-derives $\gamma_t = 3$ (rejecting all 211 smaller subsets), $\text{dist}_{\text{even}}(v) = 9$ and $\text{hor}_{\text{even}}(v) = 28$ for all 20 vertices, evaluates all six convention combinations, and runs the database-sanity gate: 989 of 994 connected graphs of order 3–7 are applicable, with 0 violations and 36 exact equalities.
- `scripts/profile_c5k4.py` and `data/profile.json` give the full certified invariant profile of $C_5[K_4]$ used throughout Section 7. (The script additionally requires `pulp/CBC` for the ILP-computed entries; the committed `profile.json` is the artefact.)
- `results/open_sweep/`, `results/refuted_sweep/`, `results/proved_sweep/` contain one JSONL verdict record per conjecture — 220, 139 and 163 records respectively — each listing every reading evaluated, the values used, and per-reading verdicts. The counts in Section 7 are obtained by tallying the `verdict` fields of these files.
- `results/family_forest.md` and `results/family_domination.md` contain the 30-member and 13-member family sweeps, the discarded-reading table for the database-sanity gate, and the slack matrices from which the sharpness lists are read.
- `results/transcription_audit.md` documents the recovery of conjectures 136–138 and the 2026-08-06 page edit; `results/verification.md` contains the adversarial re-verification of 64, 412f and 448b; `data/INVARIANT-GLOSSARY.md` contains DeLaViña’s verbatim invariant definitions recovered from `wowIIdefs.js`.
- `paper/discretization-cliff/verify_paper_claims.py` (accompanying this paper) recomputes, by exhaustive enumeration and independently of all the above, every closed form asserted in Sections 4–5 for $m = 1, \dots, 6$: α , b , f , tree, path, $\text{dist}_{\text{even}}$, hor_{even} , γ , γ_t , γ_c , L_s , $n \bmod \Delta$, $\min |N_{\overline{G}}(e)|$, and the four right-hand sides.
- Lean 4 certificates without `sorry` for the refutations are in `lean/`.

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